

1. find the eigen values and eigen vectors of following matrices

$$A = \begin{bmatrix} 0 & 5 & -10 \\ 0 & 22 & 16 \\ 0 & -9 & -2 \end{bmatrix}$$

Let $A = \begin{bmatrix} -5 & 2 \\ -7 & 4 \end{bmatrix}$. Find its eigenvalues and eigenvectors.

$$A = \begin{bmatrix} 5 & -10 & -5 \\ 2 & 14 & 2 \\ -4 & -8 & 6 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 2 & -2 \\ 1 & 3 & -1 \\ -1 & 1 & 1 \end{bmatrix}$$

2. for scaling operation along X with factor = k, what are the eigenvectors and eigen values

3. consider the following transformation.

(I) in 2D first do shear along X direction with const = k

(ii) rotate all points in XY plane anticlock by angle A

for the following case where k=1, A = 45 degrees. Find the eigen vectors and eigen values. First by graph viewing directly, then by writing the matrix and solving